

4. (a) Limestone quarrying is an important business in the UK. Quarrying creates jobs in areas where there are often limited opportunities. There is a huge demand for the products of quarrying, such as building stone and cement. Good roads and rail links are needed for transporting the products of quarrying. Thousands of people are employed in quarrying and related industries.

Many people argue against the opening of a quarry in their area despite all the benefits of limestone quarrying. Explain **two** reasons local people might use in an attempt to stop the opening of a new quarry in their area. [2]

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- (b) When limestone is heated in a lime kiln thermal decomposition takes place.

Write a balanced **symbol** equation for this reaction. [1]

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- (c) Calcium silicide is a compound containing calcium and silicon only. A sample of calcium silicide was found to contain 2.0 g of calcium and 2.8 g of silicon.

Calculate the simplest formula for this compound. You **must** show your working. [3]

Simplest formula